

Objective-Aware Projection Policies for Inverse PDE Posterior Sampling

A Two-Axis Study of Schedule and Order under Partial Observations

Hansheng Zhu
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Abstract

This report studies inverse PDE posterior sampling under sparse observations, a setting common in data assimilation and scientific sensing where uncertainty, physical validity, and runtime all matter. DiffusionPDE introduced generative PDE solving under partial observation, and Physics-Constrained Flow Matching (PCFM) introduced projection-based constraint correction during sampling. Building on these ideas, I examine which projection schedule (when to project) and projection order (how strongly to project) best trade off inverse reconstruction quality, physical consistency, runtime, and trajectory stability. The nonlinear elliptic benchmark

$$-\Delta v + \kappa v^3 = u, \quad \kappa = 50, \quad N_x = 63,$$

is used with periodic boundary conditions, where full u is recovered from sparse v observations. Relative to baseline, the best tuned setting improves u_error by **21.75%** ($1.9232 \rightarrow 1.5050$) with a clear quality-runtime tradeoff. I then extend the same protocol across additional PDE families, where guidance gains range from -9.03% (Burgers BC) to -29.84% (reaction-diffusion). Across both studies, projection policy is objective-aware rather than globally fixed. More broadly, the results show that schedule/order is a practical control knob for flow-matching and other constrained generative samplers, because correction effort can be allocated across the trajectory to match application priorities.

1 Concepts Used

MEAM 4600 concepts used include PDE modeling, inverse formulation under partial observations, residual-based numerical correction, and objective-wise tradeoff analysis. I use these concepts to structure model design, inverse inference, and objective-wise evaluation.

Core formulas:

- PDE model: $-\Delta v + 50v^3 = u$ with periodic BC.
- Inverse target: recover full u from partial v observations.
- Residual: $r(u, v) = -\Delta v + 50v^3 - u$.

- Sampling + guidance:

$$x_{k+1} = x_k + \Delta t f_\theta(x_k, t_k) - g \nabla_x \mathcal{L}_{\text{obs}}(x_k).$$

- Projection:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1}) r(x_{k+1}),$$

with first-order and Gauss-Newton variants.

2 Problem Framing

In inverse PDE settings, sparse measurements make recovery inherently ambiguous: many full fields can match the same observed subset. Posterior sampling is therefore used instead of a single deterministic predictor, so uncertainty can be represented and then guided toward physically plausible solutions.

Given that context, the research question is: *which projection schedule and projection order are best for each target objective?* The focus is on mechanism and tradeoffs, so the goal is principled comparison rather than state-of-the-art claims.

3 Method Summary

The benchmark studied is

$$-\Delta v + 50v^3 = u, \quad N_x = 63, \quad \text{periodic BC.}$$

Only 10% of v is observed, and full u is recovered. This observation mask is kept fixed across comparisons so schedule/order remains the main changing factor; I keep this fixed-mask protocol to isolate schedule/order effects cleanly.

The joint state is represented as $x = [u; v] \in \mathbb{R}^{2N_x}$, with residual

$$r(x) = -\Delta v + 50v^3 - u.$$

At each sampling step, the state first moves along the learned flow with observation guidance:

$$x_{k+1} = x_k + \Delta t f_\theta(x_k, t_k) - g \nabla_x \mathcal{L}_{\text{obs}}(x_k),$$

then optional projection is applied:

$$x_{k+1} \leftarrow x_{k+1} - P_q^{-1}(x_{k+1}) r(x_{k+1}),$$

where schedule S controls *when* projection is applied, and order $q \in \{\text{first_order}, \text{gauss_newton}\}$ controls *how* each correction is computed. These two axes are separated because timing and correction strength affect different failure modes (instability, over-correction, or weak physics enforcement).

The method is intentionally simplified: one prior family, one inverse formulation, and controlled schedule/order/guidance sweeps. This scope keeps conclusions interpretable and reproducible.

Baseline definition. Baseline means the default flow-matching model trained for 80 epochs at default width/depth, evaluated on the same projection grid. This keeps the comparison fair: gains come from tuning and projection policy, not from framework changes.

4 Metrics and Protocol

To make the evaluation logic explicit before presenting numbers, metric roles are first mapped to their exact function in the inverse task.

Table 1: Metric roles in this inverse setting.

Metric	Interpretation
<code>u_error</code>	Primary inverse target quality
<code>v_error</code>	Full-state reconstruction, incl. unobserved regions
<code>obs_error</code>	Fit on observed entries only
<code>posterior_quality</code>	Composite ranking score
<code>physical_consistency</code>	Pre-cleanup PDE consistency
<code>runtime / trajectory_stability</code>	Efficiency / robustness
<code>mmse, smse, fpd</code>	Distribution-level comparison metrics
<code>ce_ic, ce_bc, ce_cl</code>	Constraint-specific errors (N/A when inapplicable)

Evaluation uses $128 \text{ samples} \times 3 \text{ observation seeds}$ in the nonlinear main study. Observation-guidance strength is swept up to 16, then baseline is compared with a stronger checkpoint (**e200_big**: 200 epochs, larger model width/depth) under identical settings. This isolates whether gains come from training, guidance strength, or projection policy.

5 Results: Nonlinear Elliptic

5.1 Objective-wise winners

Table 2 reports the best policy for each objective under the same evaluation setup (128×3 , strong-guidance runs included).

Interpretation. The objective winners are clearly separated. The quality winner is **every_5/Gauss-Newton** with **posterior_quality=3.1364**, while the

consistency winner is **every_step/Gauss-Newton** with **physical_consistency=0.000359**. This indicates that more frequent correction helps PDE consistency, but the best composite inverse score is reached with less frequent correction (every 5 steps), likely because it avoids over-correcting intermediate samples. Speed and robustness are dominated by the no-projection baseline (**0.6086 s**, **trajectory_stability=0.0**), so one single schedule/order is not optimal for all deployment goals.

Here, “larger 200-epoch checkpoint” refers to the stronger **e200_big** model, “strong guidance” refers to observation-guidance strength $g = 16$, the projection schedule sets correction frequency (**every_step** or **every_5**), and the projection order sets the correction rule (first-order or Gauss-Newton).

Key result. Compared with the baseline best row ($u_error = 1.9232$, $v_error = 1.2944$, $obs_error = 1.1504$, $post_q = 4.3679$, $runtime = 0.7357 \text{ s}$), the tuned setup (strong guidance + first-order projection every step) reaches $u_error = 1.5050$, $v_error = 0.9120$, $obs_error = 0.8126$, $post_q = 3.2295$, $runtime = 1.3741 \text{ s}$. This corresponds to -21.75% , -29.54% , -29.36% , and -26.06% improvements, with a $+86.77\%$ runtime cost.

5.2 Best inverse row vs baseline

To evaluate inverse recovery directly, the best- u row from baseline is compared against the best- u row after tuning.

Interpretation. Best inverse rows are compared directly because u_error is the primary metric for the inverse target. Row-to-row, u_error drops by **0.4182** ($1.9232 \rightarrow 1.5050$), v_error by **0.3824** ($1.2944 \rightarrow 0.9120$), obs_error by **0.3378** ($1.1504 \rightarrow 0.8126$), and $post_q$ by **1.1384** ($4.3679 \rightarrow 3.2295$), while runtime increases by **0.6384 s** ($0.7357 \rightarrow 1.3741$). **Key takeaway.** The best inverse-reconstruction setup is **strong guidance + first-order projection at every step**, possibly because frequent lightweight corrections keep each sample close to the observation-consistent region while avoiding aggressive per-step overcorrection.

Key nonlinear conclusion. If the goal is best inverse reconstruction, project at every step with first-order correction ($u_error = 1.5050$, $v_error = 0.9120$). If the goal is best physics consistency, use Gauss-Newton-heavy correction, because curvature-aware updates reduce nonlinear PDE residual more strongly per correction; in this study, that policy achieved $physical_consistency = 0.000359$. If the goal is speed or maximum stability, skip projection ($runtime = 0.6086 \text{ s}$, $trajectory_stability = 0.0$).

6 Results: Multi-Family Extension

To test whether the nonlinear finding is specific to one PDE, the same pipeline is applied to linear elliptic, heat, reaction-diffusion, Burgers (IC/BC), and a 1D Navier-Stokes surrogate. The same evaluation logic is kept so cross-family conclusions remain comparable.

The following table reports how much inverse-target error drops as guidance strength increases from weak to strong.

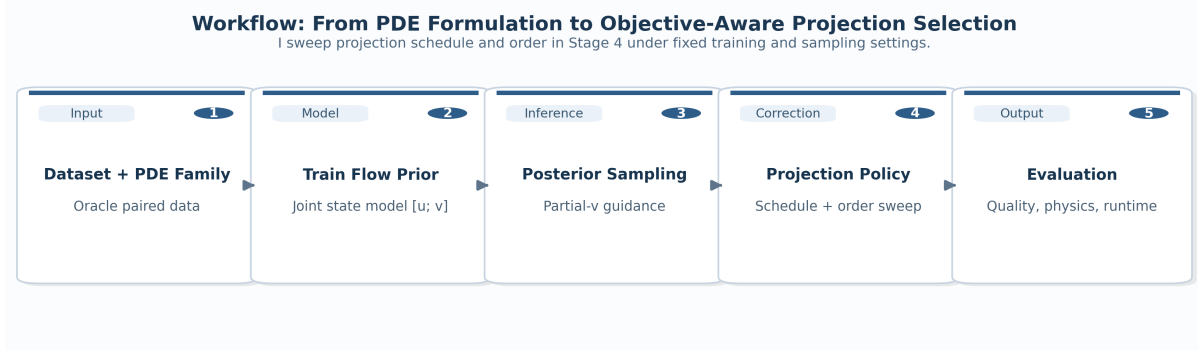


Figure 1: Workflow from PDE-family data to objective-wise schedule/order ranking. Projection is inserted inside the sampling trajectory and swept systematically.

Table 2: Best schedule/order by objective.

Objective	Winning run	Best policy	Value
posterior_quality	larger 200-epoch checkpoint + strong guidance	project every 5 steps / Gauss-Newton correction	3.1364
physical_consistency	larger 200-epoch checkpoint + strong guidance	project every step / Gauss-Newton correction	0.000359
runtime	baseline (default 80-epoch checkpoint)	no projection	0.6086 s
trajectory_stability	baseline (default 80-epoch checkpoint)	no projection	0.0

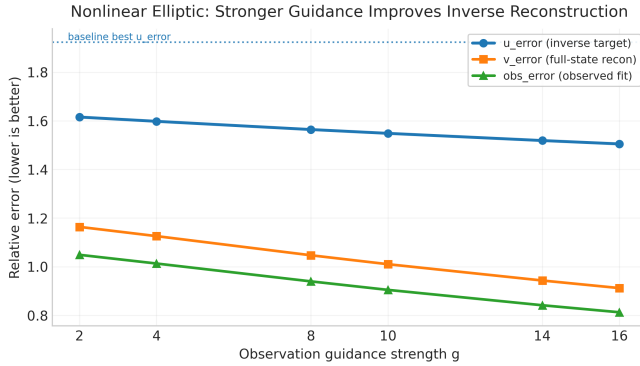


Figure 2: Stronger guidance consistently improves reconstruction metrics.

Interpretation. Across families, stronger observation guidance consistently improves inverse recovery, with u_error gains of **-18.73%**, **-25.20%**, **-29.84%**, **-26.12%**, **-9.03%**, and **-24.11%** for linear elliptic, heat, reaction-diffusion, Burgers IC, Burgers BC, and Navier-Stokes surrogate, respectively. Exact schedule/order winners remain objective-specific, matching the nonlinear benchmark.

7 Reflection and Reproducibility

What worked. Nonlinear inverse quality improved materially ($u_error : 1.9232 \rightarrow 1.5050$, $v_error : 1.2944 \rightarrow$

0.9120), and complete objective-wise ranking tables were produced.

What did not always work. Better reconstruction required higher compute (runtime $0.7357 \rightarrow 1.3741$ s), and larger/longer training was not universally better across families.

What I learned. Projection policy should be selected by objective: every-step first-order gave best inverse recovery, every-step Gauss-Newton gave best physics consistency (0.000359), and no projection gave best runtime/stability.

Attribution and responsibility. Existing repository components (especially `src/chonkdif`) were used as a numerical backend, and the posterior-projection study was implemented/extended in `src/posterior_projection`. LLM assistance was used for coding/report drafting, and metric values, commands, and final conclusions were manually verified before submission.

8 Limitations and Next Steps

The method was intentionally simplified, which improved interpretability but limits direct comparability to larger benchmark suites. In particular, full-scale reproduction of all external baselines and exhaustive architecture searches were not attempted. A natural next step is to reuse this same evaluation protocol in broader cross-method studies (using `mmse/smse/fpd` and family-specific constraint errors) while preserving the objective-wise projection analysis developed here.

Table 3: Best inverse row comparison (128×3).

Setting	u_err	v_err	obs_err	post_q	runtime
Baseline best row	1.9232	1.2944	1.1504	4.3679	0.7357 s
Best tuned (strong guidance, projection every step, first-order correction)	1.5050	0.9120	0.8126	3.2295	1.3741 s
Relative change	−21.75%	−29.54%	−29.36%	−26.06%	+86.77%

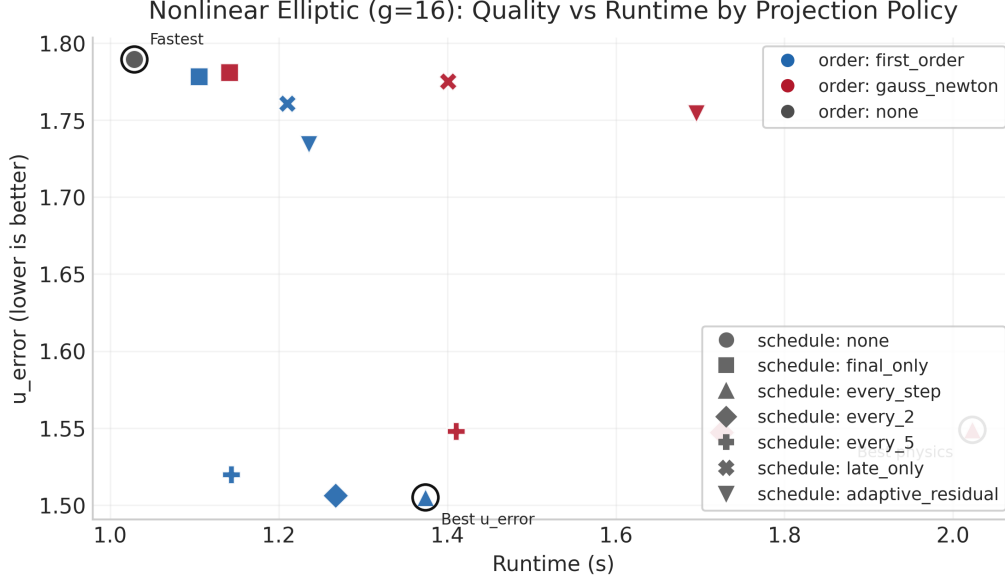


Figure 3: Quality-runtime frontier at guidance strength 16. Color shows correction rule, marker shows projection schedule, and rings mark best inverse quality, fastest runtime, and best physics consistency.

Table 4: Best-row u_error gain from $g = 0.25$ to $g = 16$.

Family	u_error reduction
Linear elliptic	−18.73%
Heat equation	−25.20%
Reaction-diffusion	−29.84%
Burgers IC	−26.12%
Burgers BC	−9.03%
Navier-Stokes 1D surrogate	−24.11%

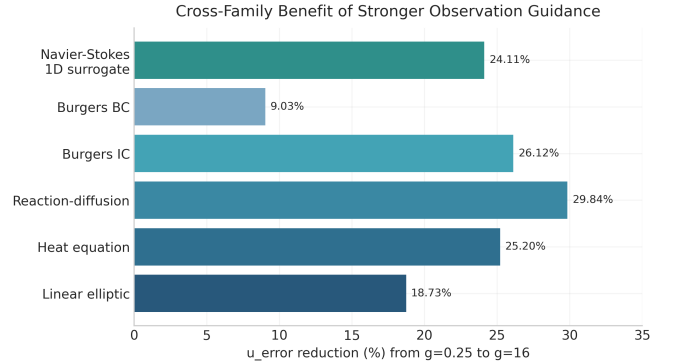


Figure 4: Cross-family trend of stronger-guidance benefit.

9 Conclusion

Projection schedule/order is a first-class algorithmic choice in partial-observation inverse PDE sampling. On the proposal benchmark, the **strongest inverse recovery** is achieved by strong guidance plus every-step first-order projection ($u_error=1.5050$, $v_error=0.9120$); **best physics consistency** is achieved by every-step Gauss-Newton projection (0.000359); and **best runtime/stability** is achieved by no projection (0.6086 s, 0.0). Therefore, projection design should match the objective of interest rather than using one fixed policy.

References

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- [2] U. Utkarsh, P. Cai, A. Edelman, R. Gomez-Bombarelli, and C. V. Rackauckas, *Physics-Constrained Flow Matching: Sampling Generative Models with Hard Constraints*, 2025.
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